

Article

Monocular Vision-Based Clamping-Point Determination via Pose Estimation for Walnut Vibration Harvesting

Ruichao Luo ¹, Xiaopeng Yang ¹, Leilei He ¹, Wulan Mao ² , Rui Li ¹, Spyros Fountas ³, Liling Yang ^{2,*} and Longsheng Fu ^{1,4,*} 

¹ College of Mechanical and Electronic Engineering, Northwest A&F University, Xianyang 712100, China; luorc@nwafu.edu.cn (R.L.); 2025056108@nwafu.edu.cn (X.Y.); hll@nwafu.edu.cn (L.H.); ruili1216@nwafu.edu.cn (R.L.)

² Institute of Agricultural Mechanization, Xinjiang Academy of Agricultural Sciences, Urumqi 830000, China; maowulan@nwafu.edu.cn

³ Agricultural University of Athens, 11855 Athens, Greece; sfountas@aua.gr

⁴ School of Electronic Information and Artificial Intelligence, West Anhui University, Lu'an 237012, China

* Correspondence: yangliling627@xaas.ac.cn (L.Y.); fulsh@nwafu.edu.cn (L.F.)

Abstract

Efficient vibration based walnut harvesting relies on the accurate determination of clamping-points on tree trunks. Properly selected clamping points can significantly enhance vibration transmission efficiency while minimizing mechanical damage to trees. However, most existing studies focus on developing generalized vibration models applicable to multiple trees, often overlooking the structural uniqueness of individual trees in clamping-point determination. This study proposes a monocular vision-based method for clamping-point determination in walnut vibration harvesting. Robustness and applicability in complex orchard environments are enhanced by introducing three keypoint annotation strategies with varying levels of structural constraints, namely 5-key-part annotation (5-KAS), 2-key-part annotation (2-KAS), and single-key-part annotation (1-KAS). Pose estimation models based on the YOLO architecture, including YOLOv8-pose, YOLO11-pose, and YOLOv12-pose, were evaluated to examine the effect of structure assisted annotation, and the results show that introducing structural constraints improves detection accuracy, training stability, and robustness. The YOLOv12-pose model combined with the 5-KAS achieves the best performance, with a precision of 95.8% and mean average precision (mAP) of 95.5%. Field harvesting experiments demonstrate that clamping-point prediction incorporating structural information achieves higher and more stable net harvesting rates. Overall, the proposed method offers a reliable and deployable solution for clamping-point determination using monocular RGB images, facilitating intelligent vibration harvesting in walnut orchards.



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Keywords: walnut vibration harvesting; clamping-point determination; pose estimation; field validation

1. Introduction

Walnut (*Juglans regia* L.) is one of the most important economic nut crops in China. Despite rapid progress in orchard mechanization, walnut harvesting still relies heavily on manual operations such as climbing, branch shaking, and stick beating [1]. It is estimated that more than 70% of walnuts are still harvested manually, which is associated with high labor intensity, low efficiency, and increased production costs, particularly in orchards

with irregular canopy structures and variable trunk morphology [2]. Similar challenges have been reported for other nut crops, including pecan, almond, and jujube, where labor shortages and rising costs further increase the need for large scale mechanized harvesting [3,4]. With the continuous decline in agricultural labor availability, manual harvesting has become increasingly difficult to sustain. Meanwhile, the expansion of orchard scale has intensified the demand for intelligent harvesting technologies in the walnut industry [5].

Vibration based harvesting has been extensively investigated for nut crops, with numerous studies focusing on vibration mechanisms, fruit detachment characteristics, and harvesting performance optimization. Previous research has examined the influence of excitation parameters, such as vibration frequency, amplitude, and duration, on fruit detachment efficiency and tree response [6]. Other studies have analyzed the dynamic behavior of tree structures during vibration harvesting, highlighting the effects of trunk stiffness, branching configuration, and resonance characteristics on vibration transmission and energy efficiency [7,8]. Despite these advances, clamping location is often treated as a predefined or manually selected parameter in existing vibration harvesting studies. In practice, inappropriate clamping positions can lead to poor vibration propagation, increased energy consumption, and excessive mechanical damage, including bark tearing, cambium abrasion, and branch breakage [9]. However, systematic methods for accurately and adaptively determining clamping-points under variable tree morphologies remain limited, particularly in field conditions. This limitation constrains the practical effectiveness and automation potential of vibration based walnut harvesting systems.

Recent advances in machine vision and deep learning have enabled reliable perception of tree structure in orchard environments, providing new opportunities for intelligent harvesting. Vision-based methods have been widely applied for trunk detection, diameter estimation, and structural segmentation using RGB images [10–12]. Lightweight convolutional neural networks, particularly YOLO based architectures, have demonstrated efficient real time performance in complex orchard scenes and have been integrated into harvesting and navigation systems [13,14]. More recently, pose estimation networks have shown promising capability in extracting structured geometric information through keypoint detection, providing an effective representation of object structure and spatial relationships [15,16]. This characteristic makes pose estimation particularly suitable for tasks that require structure aware perception and spatially constrained decision making, such as harvesting operations involving direct tool tree interaction.

Despite recent progress in pose-based agricultural perception, most existing studies focus on general structural detection, such as trunk localization or crop keypoint extraction, without explicitly applying these methods to walnut trees or clamping-point detection. Existing keypoint detection techniques are typically used to locate parts of the tree but do not leverage the overall structural information needed for clamping-point determination. In contrast, pose estimation (keypoint detection) is inherently well suited to this task because it captures the holistic structure of the tree, including key features like the trunk, branches, and intersections. Clamping-point detection, specifically for vibration harvesting, requires not just visual localization but an understanding of the tree's complete geometry and mechanical constraints. Clamping-point detection for walnut vibration harvesting remains largely unexplored [17,18]. Many vision-based studies report detection performance, but do not evaluate whether the predictions remain reliable during real harvesting operations. As a result, their performance may degrade in the presence of occlusion, background clutter, and viewpoint variation, limiting their direct applicability to vibration harvesting scenarios [19–21].

This study proposes a monocular vision-based clamping-point detection framework for vibration based walnut harvesting. In practical harvesting operations, the clamping point refers to a mechanically feasible location on the trunk that ensures efficient vibration transmission while minimizing structural damage [22,23]. Pose estimation models from the YOLO family are employed to learn structure aware representations of walnut trees and to predict clamping points based on spatial relationships between key structural components. By systematically comparing different pose models and annotation strategies, this study investigates how structural context influences clamping-point detection accuracy and harvesting performance. The proposed method relies solely on monocular RGB images, making it suitable for low cost, embedded vibration harvesting systems.

2. Materials and Methods

The proposed monocular vision-based clamping-point detection framework follows a complete workflow from image acquisition and annotation to pose-based clamping-point prediction and field validation, as illustrated in Figure 1. Monocular RGB images of walnut trees are collected from natural orchard environments to capture realistic variations in tree morphology, illumination, and background conditions. Three annotation strategies with different levels of structural information are designed to analyze the influence of structural constraints on clamping-point detection. Data processing and augmentation are then applied to enhance dataset diversity, improve model generalization, and reduce sensitivity to environmental variability. Based on the processed dataset, pose estimation models are trained and evaluated, and the predicted clamping points are finally validated through vibration harvesting experiments to assess their practical effectiveness. The detection pipeline is summarized in Algorithm 1.

Algorithm 1: Structure-assisted clamping-point detection.

Input:

Walnut tree RGB image set I

Annotation strategy $S \in \{1\text{-KAS}, 2\text{-KAS}, 5\text{-KAS}\}$

Pose model $M \in \{\text{YOLOv8-pose}, \text{YOLO11-pose}, \text{YOLOv12-pose}\}$

Output:

Predicted clamping-point coordinates P

1: Acquire orchard images I under varying viewpoints and distances

2: Annotate dataset according to selected strategy S

3: Perform data preprocessing:

- augmentation

- normalization

- dataset split

4: Initialize pose estimation model M

5: Train M using annotated dataset

6: For each test image I_i :

Extract multi-scale features

Predict keypoint coordinates P_i

7: Compute detection metrics and localization error

8: Validate P_i via vibration harvesting experiments

9: Return predicted clamping-points P

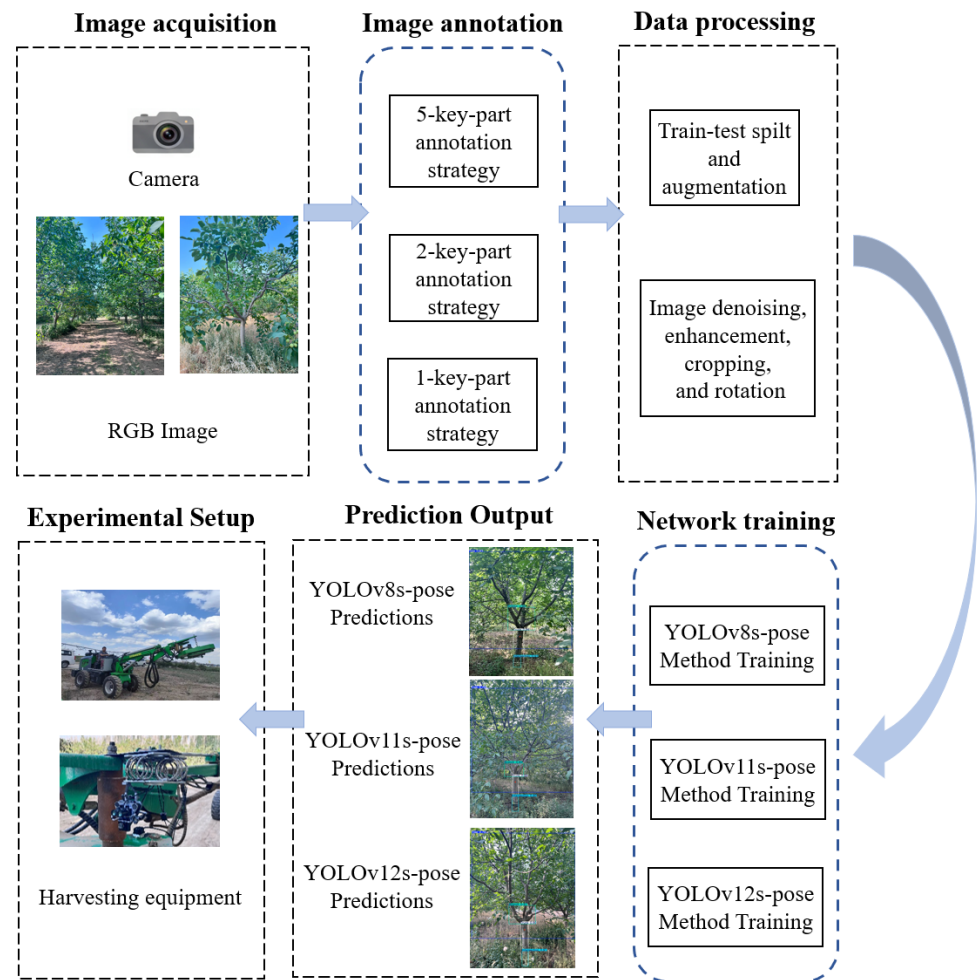


Figure 1. Overall workflow of the proposed vision-based clamping-point detection framework for walnut vibration harvesting.

2.1. Image Acquisition

Field experiments were conducted in Yecheng County, Kashgar Prefecture, Xinjiang Uygur Autonomous Region, China (77.55° E, 37.86° N), which is located at the southwestern edge of the Tarim Basin and is characterized by a temperate continental arid climate. The average elevation is 1350 m, with an annual precipitation of about 54 mm and an annual evaporation of 2100 mm. The region exhibits scarce rainfall, strong evaporation, large diurnal temperature variations, and long sunshine duration. The mean annual temperature is around 13.3 °C. These conditions represent a typical arid walnut growing environment in northwestern China.

The walnut trees used in this study were 8~12 years old and belonged to the cultivar *Wen 185*. The orchard was planted with a row spacing of approximately 8 m and a plant spacing of approximately 5 m. A total of 80 walnut trees were selected as independent experimental units. Images were captured during both the fruiting stage and the harvesting stage to ensure variability in appearance and background conditions. All images were acquired using a smartphone camera, with images captured at a native resolution of 4032 × 3024 pixels. The dataset includes multi angle full tree images as well as close up trunk images. Manual measurements of trunk diameter and trunk height (defined as the vertical distance from the trunk ground junction to the first branching point) were collected using a measuring tape for reference. Image acquisition conditions, including camera viewpoint, distance, and occlusion level, were randomized to reflect practical harvesting scenarios. All captured images were retained for subsequent annotation and analysis.

The dataset is publicly available at <https://github.com/Smart-Ag/walnut>, accessed on 1 December 2025.

2.2. Image Annotation and Data Preprocessing

Three annotation strategies with different levels of auxiliary structural information were designed to analyze the influence of structural context on clamping-point detection. For convenience, these strategies are referred to as 5-key-part annotation strategy (5-KAS), 2-key-part annotation strategy (2-KAS), and single-key-part annotation strategy (1-KAS). It should be noted that in all strategies, only one operational keypoint (clamping-point) is annotated. The difference among the strategies lies in the number of auxiliary structural bounding boxes provided during training, as shown in Figure 2. The three annotation strategies were intentionally designed to represent different levels of structural supervision rather than arbitrary labeling choices. The 5-KAS configuration provides dense structural constraints to guide clamping-point prediction within mechanically feasible trunk regions, emphasizing operational reliability. The 2-KAS configuration retains only trunk level supervision to balance annotation effort and detection stability, reflecting practical deployment considerations. The 1-KAS configuration serves as a minimal baseline to evaluate the limitations of keypoint only supervision. By comparing these structured annotation levels, the study systematically analyzes how structural context influences detection robustness and harvesting performance.

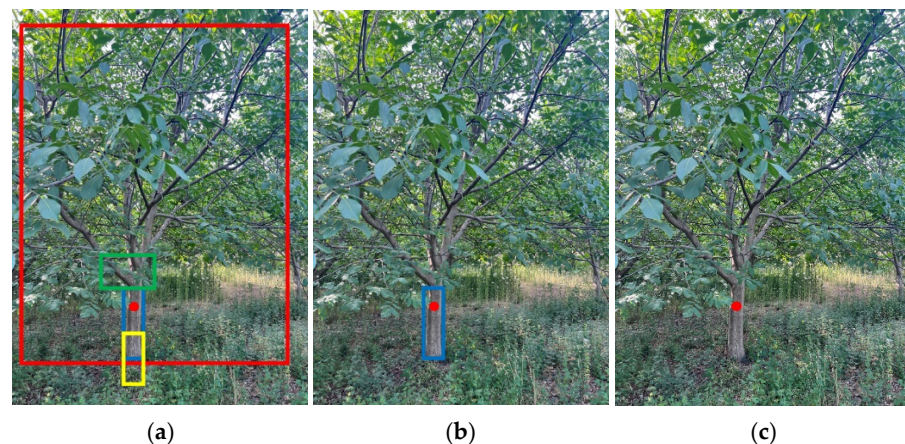


Figure 2. A visual comparison of three key part annotation strategies for clamping-point detection, including (a) the five-key-part annotation strategy (5-KAS), (b) the two-key-part annotation strategy (2-KAS), and (c) the single-key-part annotation strategy (1-KAS). In the 5-KAS strategy, the red bounding box denotes the entire walnut tree, the blue bounding box represents the trunk, the green bounding box indicates the first branching point, and the yellow bounding box corresponds to the trunk ground junction. The clamping point is annotated as a red keypoint.

In the 5-KAS strategy, four structural regions were annotated using bounding boxes, including the entire walnut tree (tree), the trunk, the first branching point of the trunk (first branch), and the trunk ground junction (trunk ground). In addition, one keypoint representing the clamping point (clamping-point) was annotated on the trunk. In the 2-KAS strategy, structural annotations were simplified to a single bounding box corresponding to the trunk, together with the clamping-point keypoint. This strategy retains basic trunk constraints while reducing annotation complexity. In the 1-KAS strategy, only the clamping-point keypoint was annotated, without any auxiliary bounding box information. This represents the minimal annotation configuration and serves as a baseline for evaluating the effect of structural constraints.

All annotations were manually performed using the LabelImg tool (<https://github.com/tzutalin/labelImg>, accessed on 1 December 2025). Bounding boxes were drawn tangent to the visible contours of the corresponding structural parts. The clamping-point annotations were carried out under the guidance of experienced field operators to ensure practical relevance and operational feasibility. Annotation files were stored in XML format, including image metadata, category labels, and pixel coordinates.

After annotation, the dataset for each strategy consisted of 374 original images. The dataset was first divided into training and validation sets with a ratio of 7:3. Data augmentation was applied to improve model robustness under varying field conditions, including horizontal flipping, random rotation, random translation, center cropping and scaling, brightness adjustment, Gaussian blur, Gaussian noise, and combined augmentation. Random rotations were performed within a range of -30° to 30° to simulate changes in camera orientation during harvesting operations. After augmentation, the dataset size increased to 3366 images in total, which includes 2356 training images and 1010 validation images.

2.3. Data Classification

The dataset was further categorized based on the horizontal distance between the camera and the tree trunk to reflect the operational workflow of vibration harvesting equipment. Three distance ranges were defined: $0.5\text{ m} < x \leq 2\text{ m}$, $2\text{ m} < x \leq 3.5\text{ m}$, and $3.5\text{ m} < x \leq 5\text{ m}$. The range of $0.5\text{ m} < x \leq 2\text{ m}$ was defined as the near view dataset, while the remaining two ranges were grouped as the far view dataset. This classification represents the visual perspectives encountered during the approach and alignment phases of trunk clamping, as shown in Figure 3. After data augmentation, the near view dataset contained 628 images, and the far view dataset contained 2738 images. This distance-based categorization was used to analyze the influence of visual scale and perspective on clamping-point detection performance.

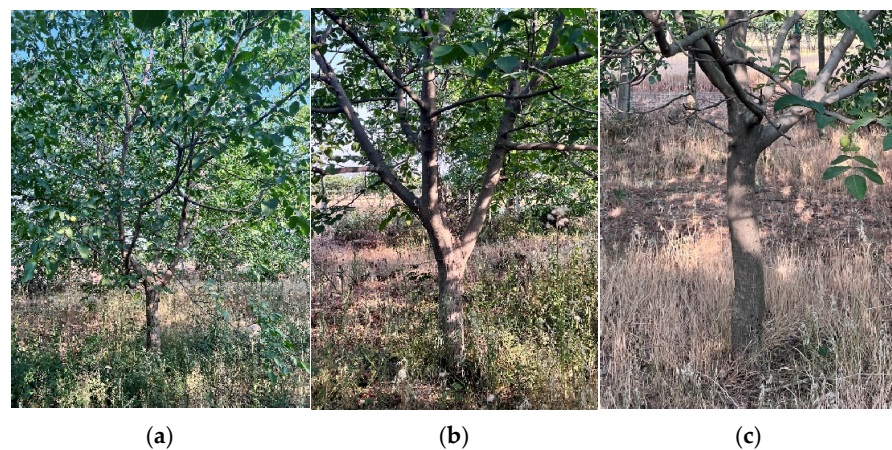


Figure 3. Examples of image views captured at different camera to tree distances of (a) far view, where $3.5\text{ m} < x \leq 5\text{ m}$; (b) far view, where $2\text{ m} < x \leq 3.5\text{ m}$; and (c) near view, where $0.5\text{ m} < x \leq 2\text{ m}$, with x denoting the distance between the camera and the tree trunk.

2.4. Model Training

Pose estimation modules within the YOLO family provide an effective approach for detecting structural keypoints, making them well suited for clamping-point detection in harvesting applications. Pose estimation modules extend conventional object detection by explicitly modeling structured keypoints associated with an object. In YOLO-based pose frameworks, the network first extracts multi-scale visual features through a convolutional backbone and feature aggregation layers. These features encode both local appearance cues and global structural context of the walnut tree. A dedicated pose head then regresses

normalized coordinates of predefined keypoints by learning spatial relationships within the trunk region. Unlike pure detection tasks that focus only on bounding boxes, pose estimation integrates structural priors into the prediction process, enabling more stable localization of mechanically meaningful points. In the context of walnut vibration harvesting, this structure aware mechanism allows the model to interpret trunk geometry and constrain the predicted clamping-point to physically feasible regions, thereby improving robustness under occlusion, illumination variation, and morphological diversity.

Transformer-based pose estimation frameworks have demonstrated strong global context modeling capability in generic vision tasks. However, vibration harvesting systems impose strict constraints on real time inference speed, computational efficiency, and deployability on embedded platforms. YOLO-based pose architectures offer a favorable balance between detection accuracy and computational cost, enabling stable performance under moderate dataset scales typical of agricultural applications. Furthermore, clamping-point detection primarily relies on localized trunk structure and spatial constraints rather than long range global reasoning. Under such task characteristics, convolution-based pose models provide sufficient representational capacity while maintaining operational efficiency. Therefore, YOLO-based pose models were selected as a practical and task-oriented solution for clamping-point detection in vibration harvesting.

In this study, three pose detection models of YOLOv8-pose, YOLO11-pose, and YOLOv12-pose variants were trained and evaluated to examine their capability for walnut clamping-point detection, respectively. These models differ in backbone configurations, feature aggregation mechanisms, and keypoint prediction strategies, which may affect their performance under complex orchard environments. By conducting a comparative analysis across the three models, this study aims to assess their robustness, keypoint detection accuracy, and overall detection reliability when applied to trunk related pose estimation. The benchmarking results provide a solid basis for selecting the most suitable pose detection architecture to support accurate and stable clamping-point determination in vibration harvesting operations.

The experimental environment for this study was configured on a 64-bit Windows 10 operating system equipped with 128 GB RAM, an Intel Xeon Gold 5128 32-core processor (2.3 GHz), and dual NVIDIA GeForce RTX 4090 graphics cards (24 GB). The deep learning framework used was PyTorch 2.0.1, with CUDA 12.1 and cuDNN 8.8.1, as shown in Table 1. Additional software dependencies included Python 3.9 and OpenCV 4.8. The detailed hardware and software environment is summarized in Table 1. The model input size was set to 640×640 pixels with a batch size of 16. Training was performed using stochastic gradient descent with a momentum of 0.937 and a weight decay of 0.0005. The initial learning rate was set to 0.01, and the training process was conducted for 200 epochs.

Table 1. Model training parameters.

Items	Parameters
CPU	Intel Xeon Gold 5128 32-core processor (2.3 GHz)
RAM	
Operating system	64-bit Windows 10
GPU	dual Nvidia GeForce RTX 4090 24G GPUs
Development	Python 3.9, PyTorch 2.0.1
Environments	CUDA 12.1

2.5. Performance Evaluation

Since clamping-point detection combines object level localization and keypoint level accuracy, two complementary evaluation perspectives were adopted in this study. De-

tection oriented metrics, including Precision, Recall, AP, and mAP, were used to assess whether clamping-points could be reliably detected under different annotation strategies and model architectures. In addition, a distance-based keypoint error metric was introduced to quantitatively evaluate the positional accuracy of the predicted clamping-point relative to the operator annotated ground truth.

Model performance was evaluated using Precision (P), Recall (R), Average Precision (AP), and mean Average Precision (mAP). These metrics are widely applied in pose estimation tasks. A true positive (TP) is defined as a predicted clamping-point located within a predefined distance threshold from the ground truth. False positives (FP) indicate incorrect predictions, and false negatives (FN) represent missed clamping-points. Precision and Recall were calculated based on TP , FP , and FN . AP was computed as the area under the Precision Recall curve, and mAP was obtained by averaging AP values over all clamping-point categories. These metrics jointly assess the accuracy and reliability of clamping-point detection, defined as shown in Equations (1)–(4).

$$P = \frac{TP}{TP + FP} \quad (1)$$

$$R = \frac{TP}{TP + FN} \quad (2)$$

$$AP = \int_0^1 P(R) dR \quad (3)$$

$$mAP = \frac{1}{n} \sum_{i=1}^n AP_i \quad (4)$$

In the clamping-point detection experiment, the evaluation metric was defined as the pixel Euclidean distance between the predicted clamping-point and the ground truth clamping-point. Let the normalized coordinates of the ground truth clamping-point be (x, y) , and those of the predicted clamping-point be (x', y') . Given an image resolution of $w \times h$, Equation (5) calculates the pixel distance between the two points in the horizontal direction. Equation (6) represents the pixel distance in the vertical direction, and Equation (7) defines the pixel Euclidean distance between the two points.

$$D_x = w(x' - x) \quad (5)$$

$$D_y = h(y' - y) \quad (6)$$

$$O = \sqrt{D_x^2 + D_y^2} \quad (7)$$

where x and y denote the normalized horizontal and vertical coordinates of the ground truth clamping-point, respectively; x' and y' represent the normalized horizontal and vertical coordinates of the predicted clamping-point, respectively. w and h indicate the width and height of the input image in pixels, respectively. D_x and D_y are the pixel distances between the predicted and ground truth clamping-points along the horizontal and vertical directions, respectively. O denotes the pixel Euclidean distance between the two clamping-points.

2.6. Experimental Setup

The walnut vibration harvester used in this study was developed by the Institute of Agricultural Equipment, Xinjiang Academy of Agricultural Sciences, as shown in Figure 4a. The harvester was modified based on a John Deere 1204 tractor and equipped with a dedicated end effector. The main parameters of the end effector are listed in Table 1. A camera and a vibration damping platform were mounted on the end effector, as illustrated in Figure 4b.



Figure 4. The experimental platform for field tests includes (a) a walnut vibration harvester and (b) a camera and gimbal mounting setup.

Walnut trees aged 8~12 years with similar morphology were selected for the experiments. Vibration harvesting tests were conducted at the clamping-point and at positions 10 cm above and below the clamping-point. During each test, the detached walnuts were collected and weighed. After machine harvesting, the remaining walnuts were manually harvested and weighed. The mass of walnuts harvested by the machine is denoted as m , and the mass collected manually is denoted as p . The fruit removal rate was calculated as Equation (8).

$$R_f = \frac{m}{m + p} \quad (8)$$

where R_f denotes the net harvesting rates; m represents the mass of walnuts detached and harvested by the vibration harvester during the test; p denotes the mass of walnuts remaining on the tree and subsequently collected manually after machine harvesting. Both m and p are measured in kilograms.

3. Results and Discussion

3.1. Comparison of Different Keypoint Annotation Strategies

Structural information embedded in annotation has a decisive impact on clamping-point detection performance. Across all pose estimation models, a clear and consistent performance hierarchy is observed among the three KAS configurations. As shown in Table 2, 5-KAS achieves the highest detection accuracy for all models. In particular, YOLOv12-pose with 5-KAS reaches a Precision of 95.8% and an mAP@0.5 of 95.5%, outperforming the corresponding 2-KAS and 1-KAS configurations by 3.1~8.0 percentage points in mAP. These results indicate that dense structural cues effectively constrain clamping-point predictions within mechanically feasible trunk regions.

Beyond accuracy improvement, 5-KAS enhances the stability and reliability of clamping-point detection. By explicitly encoding the spatial relationships among the trunk, first branching position, and trunk ground intersection, the model is guided to learn structure aware representations rather than relying solely on local appearance cues. This leads to more consistent predictions under background clutter and partial occlusion, which are common in orchard environments, as shown in Table 2. The 2-KAS configuration exhibits a meaningful balance between performance and annotation cost. Although its accuracy is slightly lower than that of 5-KAS, 2-KAS maintains stable detection performance, with mAP@0.5 values remaining above 88.9% for all models. This suggests that trunk level structural constraints alone are sufficient to restrict predictions to a reasonable operational range, making 2-KAS a practical choice for large scale deployment where annotation efficiency is critical.

Table 2. Detection performance (Precision, Recall, mAP) of the nine models formed by three pose models and three annotation strategies.

Model	Strategy	Precision (%)	Recall (%)	mAP@0.5 (%)
YOLOv8-pose	1-KAS	82.4	79.1	81.5
	2-KAS	86.5	83.2	85.1
	5-KAS	89.2	87.4	88.6
YOLO11-pose	1-KAS	85.1	82.3	84.4
	2-KAS	89.7	86.8	88.9
	5-KAS	92.4	91.2	92.0
YOLOv12-pose	1-KAS	88.3	85.6	87.5
	2-KAS	93.1	90.5	92.4
	5-KAS	95.8	94.2	95.5

Note: The five-key-part annotation strategy (5-KAS) includes four structural bounding boxes (tree, trunk, first branching point, and trunk ground intersection) and one clamping-point keypoint. The two-key-part annotation strategy (2-KAS) consists of one trunk bounding box and one clamping-point keypoint. The single-key-part annotation strategy (1-KAS) annotates only the clamping-point keypoint without any bounding box constraints.

In contrast, the degradation observed with 1-KAS highlights the limitations of keypoint only supervision. Without explicit structural boundaries, detection performance drops by up to 8.0% in mAP@0.5 compared with 5-KAS, and predictions become more sensitive to trunk morphology variation and background interference. Qualitative results further show that 1-KAS produces less stable spatial distributions, with clamping-points occasionally deviating toward visually similar but mechanically unreasonable regions. These findings confirm that explicit structural context is essential for reliable clamping-point detection in vibration harvesting applications.

Overall, the proposed KAS framework represents a shift from appearance driven detection toward operation-oriented perception. By embedding task specific mechanical feasibility directly into annotation design, structure assisted KAS enables pose estimation models to produce predictions that are not only accurate but also operationally reliable, which is critical for real world vibration harvesting.

3.2. Performance of Pose-Based Clamping-Point Prediction

Pose-based vision models provide an effective representation for clamping-point prediction in orchard environments by explicitly modeling structural information through keypoints. Across the evaluated models, all three YOLO-based pose estimation architectures are able to detect trunk located clamping points under natural orchard conditions, indicating that pose estimation is a suitable paradigm for structure related perception in agricultural applications, as shown in Figure 5. Similar observations have been reported in studies on tree and crop perception, where keypoint based representations were shown to facilitate robust extraction of structural information under field conditions [9,18].

Detection performance, however, varies systematically with model architecture and feature representation capacity. YOLOv8-pose achieves moderate detection performance in scenes with limited occlusion, but its Precision and Recall decrease under complex backgrounds and dense foliage, reflecting limited robustness to environmental interference. Comparable behavior has been observed in earlier lightweight pose or detection models applied to orchard scenarios, where restricted contextual modeling leads to missed or unstable predictions [20]. In contrast, YOLO11-pose exhibits improved detection Precision and Recall across different annotation settings, benefiting from enhanced feature representation and more effective multi scale information integration.

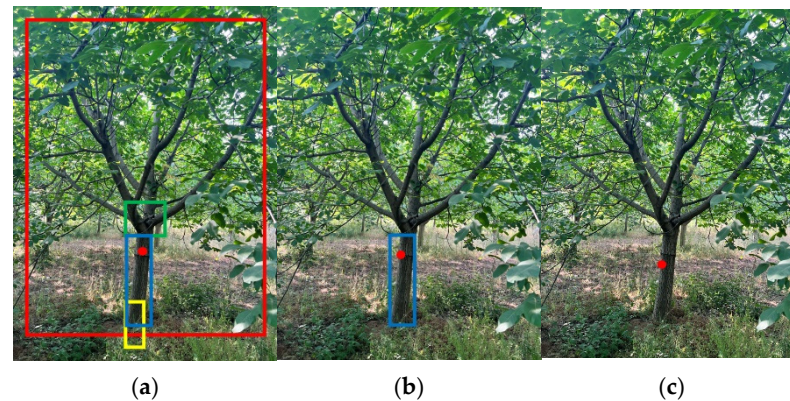


Figure 5. Visual comparison of clamping-point prediction results obtained using the same pose-estimation model under the three different key-part annotation strategies of (a) 5-key-part annotation strategy (5-KAS), (b) 2-key-part annotation strategy (2-KAS), and (c) single-key-part annotation strategy (1-KAS). In the 5-KAS strategy, the red bounding box denotes the entire walnut tree, the blue bounding box represents the trunk, the green bounding box indicates the first branching point, and the yellow bounding box corresponds to the trunk ground junction. The clamping point is annotated as a red keypoint.

YOLOv12-pose further improves detection robustness by more effectively leveraging structural context across scales. Compared with earlier architectures, YOLOv12-pose maintains higher Precision, Recall, and mAP@0.5 across all KAS configurations, and shows reduced performance degradation when structural supervision is limited, as shown in Table 2. This indicates that deeper feature hierarchies and advanced multi scale fusion enable more reliable modeling of trunk geometry and spatial relationships. YOLO-based fruit or trunk detection systems typically report precision or mAP values in the range of approximately 85%~93% under field conditions, with noticeable degradation when occlusion, background clutter, or scale variation is present. In particular, severe trunk occlusion can significantly impair detection reliability. For example, in orchards where trees have not been properly pruned, dense canopy growth may completely obscure large portions of the trunk, leaving only fragmented or visually ambiguous trunk regions visible to the camera. Under such conditions, appearance driven detection models may fail to localize the trunk correctly, leading to missed detections or unstable predictions. Recent pose estimation approaches applied to agricultural targets demonstrate improved structural awareness; however, their robustness can still be sensitive to extreme visual complexity, especially when key structural cues are partially or fully hidden [21]. Similar trends have been reported in recent agricultural vision studies, where advanced pose estimation architectures demonstrated improved robustness in complex field environments [14]. Overall, these results highlight that the effectiveness of pose-based clamping-point prediction depends not only on the pose estimation paradigm itself, but also on the ability of the underlying architecture to exploit structural cues under realistic orchard conditions.

3.2.1. Overall Performance Analysis

Clamping-point detection performance is jointly determined by the amount of structural supervision and the representational capacity of the pose-model architecture. The consistent improvement observed from 1-KAS to 5-KAS indicates that explicit structural constraints provide effective spatial priors, enabling the model to suppress ambiguous background responses and focus on mechanically feasible trunk regions. Meanwhile, the progressive performance gains from YOLOv8-pose to YOLOv12-pose suggest that advanced architectures are better suited to exploit such structural information through stronger feature representation and cross scale context integration.

The superior performance of YOLOv12-pose combined with 5-KAS arises from the complementary interaction between architecture and annotation design. Richer structural annotation defines a constrained solution space for clamping-point prediction, while the deeper architecture of YOLOv12-pose, equipped with enhanced multi scale feature fusion and attention mechanisms, effectively models long range spatial dependencies along the trunk. This synergy allows the model to maintain high Precision and Recall simultaneously, leading to a Precision of 95.8% under the 5-KAS configuration, which satisfies the accuracy requirements for autonomous vibration harvesting.

In contrast, weaker performance observed under simplified configurations can be attributed to insufficient structural guidance and limited contextual modeling. Under 1-KAS, the absence of explicit structural boundaries increases reliance on local appearance cues, making predictions more vulnerable to background clutter, trunk texture variation, and partial occlusion. When coupled with earlier architectures such as YOLOv8-pose, which have weaker global context modeling capability, this limitation becomes more pronounced, resulting in missed detections and unstable localization.

Compared with human pose estimation tasks, pose-based perception of agricultural plants poses distinct challenges that further amplify the importance of structure aware design. Human pose estimation benefits from relatively consistent skeletal topology and limited inter instance variability, whereas walnut trees exhibit significant variations in trunk diameter, branching height, inclination, and canopy structure. In addition, plant perception is more severely affected by self-occlusion, foliage interference, and background similarity. These factors reduce the effectiveness of appearance driven keypoint detection and highlight why agricultural pose estimation tasks require stronger structural priors and more robust architectural support than human centered applications.

3.2.2. Impact of KAS

Increasing the level of structural information in annotation leads to consistent improvements in clamping-point detection accuracy across all pose models. This effect is most evident for YOLOv12-pose, where the mAP@0.5 increases from 87.5% with 1-KAS to 95.5% with 5-KAS, representing an absolute improvement of 8.0%. This result indicates that while keypoint only annotation provides a baseline for clamping-point detection, the absence of explicit structural boundaries limits localization reliability. The introduction of multi-level structural constraints in 5-KAS effectively restricts the feasible prediction space, resulting in more accurate and stable clamping-point localization.

3.2.3. Comparison of Model Architectures

Differences in pose model architecture lead to distinct feature representation capabilities for clamping-point detection. Across the three generations of YOLO-pose models, YOLOv12-pose consistently demonstrates superior performance, particularly under the 5-KAS configuration, where it outperforms YOLOv8-pose by 6.9% in mAP@0.5. This performance gap indicates that the improvement arises from architectural advances in feature extraction and multi scale representation, rather than simple model scaling.

The limitations of earlier architectures become evident under complex orchard conditions. YOLOv8-pose relies on relatively shallow feature hierarchies and limited multi-scale fusion, which restrict its ability to capture long range spatial context along the trunk. As a result, although YOLOv8-pose achieves acceptable precision in visually simple scenes, its recall decreases to 87.4% under the 5-KAS configuration, indicating frequent missed detections when foliage occlusion and background clutter are present. These failures suggest that local appearance cues alone are insufficient for reliable clamping-point detection in realistic orchard environments.

In contrast, the architectural design of YOLOv12-pose better aligns with the structural characteristics of walnut trees. By incorporating enhanced multi scale feature fusion and attention mechanisms, YOLOv12-pose is able to integrate fine grained local features with global structural context along the trunk. This capability allows the model to maintain a high recall of 94.2% under the same conditions, ensuring stable clamping-point detection even when trunk boundaries are partially obscured. From an architectural perspective, this indicates that deeper feature representations and explicit attention to spatial dependencies are critical for perception tasks that require structure aware localization.

These observations confirm that model capacity and architectural design are decisive factors for pose-based clamping-point detection in field environments. While pose estimation frameworks are generally suitable for representing clamping points, their practical effectiveness depends strongly on the ability of the underlying architecture to model tree structure across scales. Advanced architectures such as YOLOv12-pose therefore provide a more reliable foundation for vibration harvesting applications operating under complex orchard conditions.

3.2.4. Training Convergence Behavior Analysis

Both annotation design and pose model architecture have a clear influence on training convergence behavior. To ensure a fair comparison across different model architectures and loss scales, the pose loss of each configuration was normalized by its initial value at epoch 10. Only the stable training stage from epoch 10 to 200 was considered to exclude early warm up effects, as shown in Figure 6. Although all models were trained with the same maximum number of epochs, some configurations reached convergence earlier or stopped logging loss values after the loss plateau was achieved. As a result, the lengths of the loss curves differ slightly. The convergence analysis therefore focuses on the common effective training interval where reliable loss records are available for all configurations.

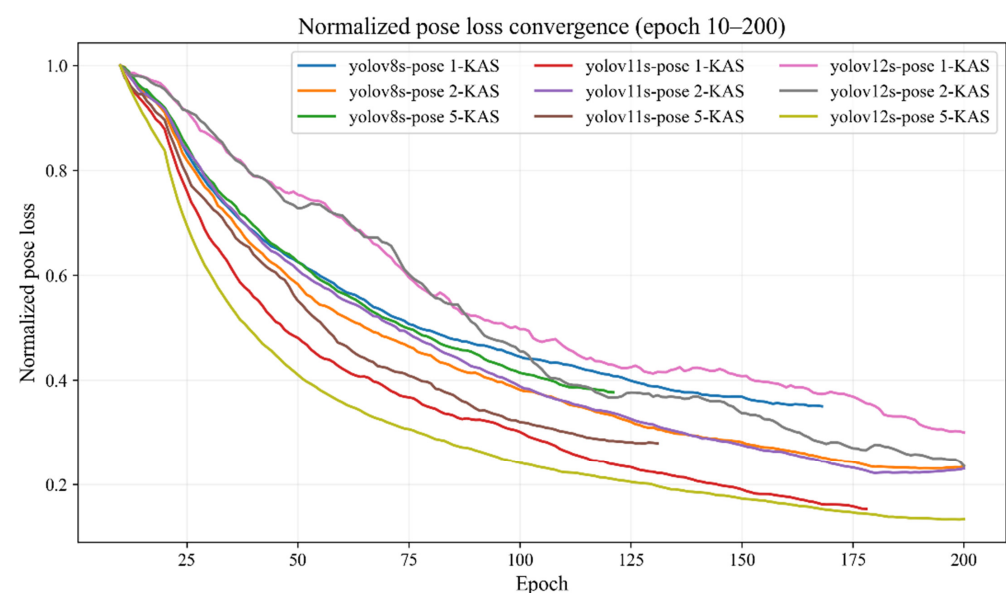


Figure 6. Normalized pose loss convergence of nine model annotation configurations during training (epoch 10–200), 5-key-part annotation strategy (5-KAS), 2-key-part annotation strategy (2-KAS), and single-key-part annotation strategy (1-KAS).

Increasing the amount of explicit structural supervision leads to faster convergence and lower final training loss across all pose models. For each model architecture, the 5-KAS configuration consistently converges more rapidly and reaches the lowest normalized loss level, followed by 2-KAS and 1-KAS. This behavior can be attributed to the progressive

introduction of structural bounding boxes that explicitly constrain the feasible region for clamping-point prediction. In particular, the inclusion of the trunk region, first branching position, and trunk round intersection in 5-KAS provides clear upper and lower spatial boundaries for keypoint learning. This structure derived spatial priors reduce ambiguity during optimization, stabilize gradient updates, and accelerate convergence. As a result, the observed convergence behavior is consistent with the superior detection accuracy achieved by structure assisted KAS configurations.

Advances in pose model architecture further enhance optimization efficiency and convergence stability. Among the three evaluated models, YOLOv12-pose exhibits the steepest loss reduction and the lowest normalized loss plateau, reflecting stronger feature representation and more efficient optimization. YOLOv11-pose shows intermediate convergence behavior, whereas YOLOv8-pose converges more slowly and remains at a higher loss level throughout training. Similar trends have been reported in studies on agricultural and outdoor pose estimation, where deeper architectures with enhanced feature fusion lead to faster convergence and more stable training under complex backgrounds [16].

These differences are consistent with the overall detection performance trends, indicating that both model capacity and annotation design influence training behavior and final detection quality in pose-based clamping-point detection. Recent agricultural vision studies have shown that structure aware supervision helps reduce training instability caused by visual clutter and scale variation by providing clearer spatial constraints [10,11]. Therefore, the use of structural bounding box constraints is particularly beneficial for achieving stable and reliable training in complex orchard environments.

3.3. Effect of Bounding Box Constraints on Clamping-Point Detection

Bounding box constraints play a critical role in reducing ambiguity in clamping-point detection by introducing explicit structural priors. The incorporation of auxiliary structural regions encourages the pose estimation model to learn not only local visual features but also the spatial topology of the walnut tree relevant to clamping-point prediction, as shown in Figure 7. These structure aware priors effectively restrict the feasible prediction region to locations that are both visually consistent and mechanically plausible, thereby improving detection reliability.

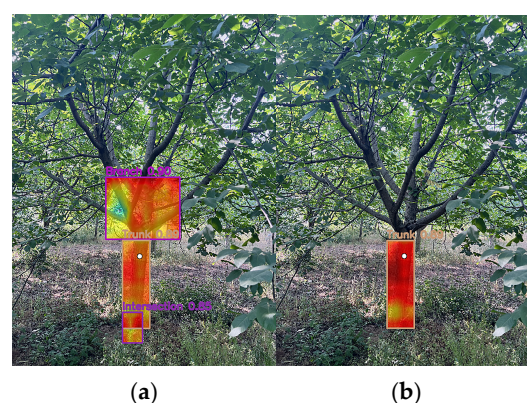


Figure 7. Heatmap visualization of clamping-point prediction results under different annotation strategies of (a) 5-KAS strategy and (b) 2-KAS strategy.

The degree of structural constraint directly affects the spatial distribution and concentration of clamping-point confidence. When only trunk level supervision is provided under the 2-KAS configuration, the predicted confidence distribution of the clamping point is relatively dispersed along the vertical direction of the trunk. Although the trunk constraint suppresses false predictions in background regions, increased uncertainty remains near the

upper trunk, particularly close to the branching area. In contrast, the 5-KAS configuration produces a more compact and vertically constrained confidence distribution. By explicitly introducing the first branching point and the trunk ground intersection as additional structural boundaries, the model learns clearer upper and lower limits for feasible clamping locations. As a result, high confidence predictions are concentrated within a narrower trunk segment that aligns well with operator experience and vibration harvesting mechanics.

Structure aware supervision improves both prediction consistency and detection robustness. Rather than relying solely on appearance-based cues, the model learns spatial relationships between the trunk and its key structural components through bounding box constraints, which significantly reduces prediction ambiguity. This mechanism explains the improved detection accuracy and stability observed for structure assisted KAS configurations in both quantitative evaluations and field harvesting experiments.

3.4. Clamping-Point Detection Accuracy Analysis

Both pose model architecture and structural constraints have a decisive impact on clamping-point localization accuracy. Clamping-point detection accuracy was quantitatively evaluated using the Euclidean distance between the predicted clamping point and the corresponding ground truth annotation. For each pose estimation model, 200 detected clamping points were randomly sampled, and the average localization error together with its statistical distribution was analyzed. Among the three pose models, YOLOv12-pose consistently achieves the smallest detection errors across all KAS configurations, followed by YOLOv11-pose and YOLOv8-pose.

This trend indicates that improved feature representation and multi scale fusion in more advanced pose models contribute directly to higher keypoint localization accuracy, as shown in Figure 8. Similar observations have been reported in previous pose estimation and agricultural vision studies, where deeper architectures and enhanced feature fusion mechanisms led to reduced keypoint localization error under complex field conditions [9,16]. In contrast, earlier or lightweight models tend to exhibit larger localization errors and reduced robustness in the presence of background clutter, occlusion, and scale variation, a tendency also reported in orchard and outdoor perception studies [20].

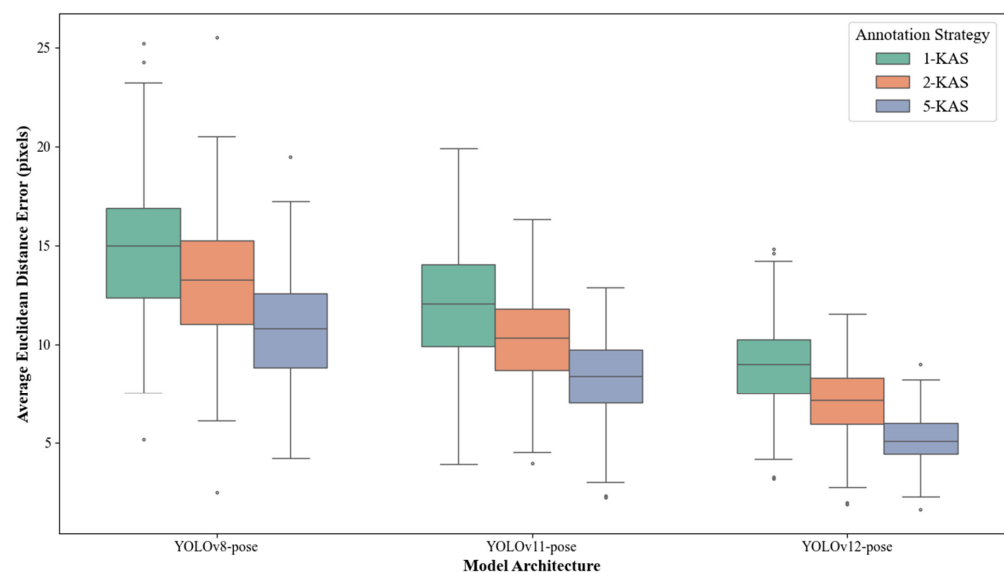


Figure 8. Distribution of clamping-point detection errors under different KAS strategies. The points in the box plot are outliers.

Richer structural constraints significantly reduce both average error and error dispersion in clamping-point detection. Across all three pose estimation models, the 5-KAS configuration results in the lowest mean localization error and the narrowest interquartile range (IQR), indicating both higher accuracy and greater prediction stability. The 2-KAS configuration shows slightly higher errors, while the 1-KAS configuration produces the largest deviations and the widest error distributions, as shown in Figure 8. This pattern demonstrates that bounding box based structural constraints effectively suppress both systematic bias and random variation by restricting predictions to mechanically plausible regions on the trunk.

The combination of advanced pose models and structure assisted KAS yields the most stable and reliable predictions. In particular, YOLOv12-pose combined with 5-KAS achieves the smallest median error and the most compact error distribution. Compared with the YOLOv8-pose model trained under the 1-KAS configuration, the YOLOv12-pose and 5-KAS configuration exhibits a substantial reduction in IQR, reflecting markedly improved prediction consistency. This improvement is especially critical for vibration harvesting applications, where small localization errors can result in suboptimal clamping positions and increased mechanical risk. Overall, these results confirm that bounding box based structural constraints not only improve average detection accuracy but also enhance the robustness and reliability of clamping-point prediction under varying orchard conditions.

3.5. Field Validation and Discussion

3.5.1. Field Experimental Setup and Evaluation Objective

Field experiments confirm that vision-based clamping-point prediction can be effectively translated into stable harvesting performance under real operating conditions. Field tests were conducted during actual vibration harvesting in September 2025 to evaluate the practical effectiveness of the proposed method, as illustrated in Figure 9. All experiments were performed using the YOLOv12s-pose model, which provides a favorable balance between detection accuracy and real time performance for embedded harvesting platforms. A total of 36 walnut trees were evaluated and divided into three groups corresponding to the 5-KAS, 2-KAS, and 1-KAS configurations. Rather than focusing solely on visual detection accuracy, the field evaluation emphasizes whether different KAS configurations lead to stable and effective harvesting outcomes when deployed on real vibration harvesting equipment. Accordingly, the net harvesting rate and its variability were selected as the primary evaluation indicators.



Figure 9. Field experimental scene of walnut vibration harvesting in an orchard.

3.5.2. Comparison of Harvesting Performance Under Different Annotation Strategies

Structure assisted KAS configurations lead to higher and more stable harvesting performance in field conditions. Field harvesting statistics indicate that 5-KAS achieves the highest mean net harvesting rate of 92.07% together with the lowest standard deviation of $\pm 4.63\%$, reflecting both high efficiency and strong stability. In comparison, 2-KAS results in a lower mean net harvesting rate of 88.49% and exhibits greater variability, suggesting that reduced structural information leads to less consistent clamping-point selection. The 1-KAS configuration attains a relatively high mean net harvesting rate of 91.66%; however, its variability remains comparable to that of 5-KAS, indicating acceptable average performance but reduced robustness, as shown in Table 3. These results indicate that keypoint only annotation can support harvesting in favorable cases, but the lack of explicit structural constraints leads to increased variability in harvesting performance when tree morphology differs across individuals. It is important to interpret the reported detection accuracy in the context of dataset scale and generalization capability. While the dataset is not large compared with generic vision benchmarks, targeted augmentation strategies and cross-validation ensured that the models learned structural relationships rather than memorizing specific samples. Field validation further confirmed that the predictions remain operationally stable under real harvesting conditions.

Table 3. Field harvesting performance under different annotation strategies.

Annotation Strategy	Detached Fruit (kg)	Remaining Fruit (kg)	Total Fruit (kg)	Net Harvesting Rate (%)
5-KAS	26.9 $\pm$ 12.1	2.4 $\pm$ 1.8	29.3 $\pm$ 13.0	92.1 $\pm$ 4.6
2-KAS	45.1 $\pm$ 7.6	6.2 $\pm$ 1.9	51.3 $\pm$ 8.4	88.5 $\pm$ 3.3
1-KAS	31.2 $\pm$ 9.4	3.6 $\pm$ 2.1	34.8 $\pm$ 10.2	91.7 $\pm$ 4.3

3.5.3. Stability Analysis Based on Net Harvesting Rate Distribution

Structural constraints have a decisive influence on the stability of harvesting performance across different trees. The distribution of net harvesting rates shows that the 5-KAS configuration achieves a higher median value together with a narrower interquartile range, indicating consistent harvesting performance under varying tree conditions. The limited dispersion of the distribution suggests that clamping-point predictions produced by 5-KAS are less sensitive to individual tree differences. In contrast, both 2-KAS and 1-KAS exhibit wider interquartile ranges and longer distribution tails, reflecting increased variability in harvesting outcomes, as shown in Figure 10. This increased variability indicates that when structural constraints are reduced, clamping-point detection becomes more sensitive to variations in trunk shape, branching pattern, and local visual conditions. As a result, harvesting performance becomes less predictable, highlighting the importance of structure aware constraints for achieving stable field operation.

3.5.4. Influence of Tree Morphology, Illumination, and Foliage Occlusion

Tree morphology and environmental conditions jointly influence the reliability of clamping-point prediction in field operations. Field observations show that trees with relatively straight trunks, clearly identifiable first branching points, and regular structural geometry tend to achieve higher and more stable net harvesting rates, particularly under 5-KAS. In these cases, structural bounding boxes provide accurate spatial references that facilitate robust keypoint detection. Similar observations have been reported in previous orchard robotics and agricultural vision studies, where regular trunk geometry and clear structural features were found to improve perception stability and downstream manipulation performance [24]. In contrast, trees with leaning trunks, irregular branching patterns,

or asymmetric canopies exhibit increased prediction uncertainty, especially when simplified KAS configurations are used, a limitation that has also been noted in structure-agnostic detection approaches [20].

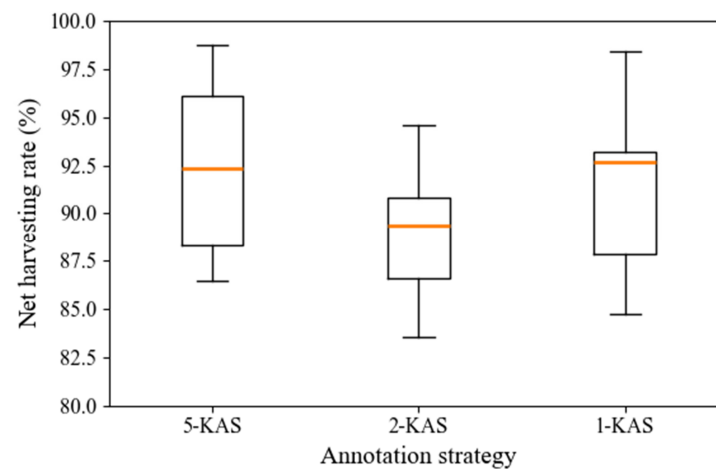


Figure 10. Comparison of net harvesting rate under different annotation strategies. The orange line in the box plot represents the mean value.

Adverse visual conditions further amplify prediction uncertainty, while structure-assisted KAS mitigates their impact. Variations in illumination, such as strong sunlight or shadowed trunk regions, together with foliage occlusion, can obscure trunk contours and degrade local visual features, leading to less reliable clamping-point detection. These challenges are well documented in agricultural vision literature, where illumination variation and occlusion are recognized as major sources of perception degradation in orchard environments [10,21]. Under such conditions, 5-KAS demonstrates stronger robustness by leveraging multiple structural cues to compensate for local visual degradation. This observation is consistent with recent findings showing that integrating explicit structural context improves robustness under adverse visual conditions and supports stable perception in real field operations [18].

3.5.5. Engineering Implications and Deployment Considerations

Structural context is a key determinant of reliable and stable clamping-point prediction in practical vibration harvesting systems. From an engineering perspective, the field results demonstrate that incorporating structural information significantly enhances harvesting stability, which is consistent with the effectiveness of pose and keypoint based perception strategies reported in automated fruit and stalk detection systems [25–28]. Although 5-KAS requires greater annotation effort, it delivers the most reliable harvesting performance and is well suited for fully autonomous systems operating under diverse orchard conditions. The 2-KAS configuration provides a practical balance between annotation cost and detection performance and may be more suitable for large scale deployment scenarios where rapid dataset construction is required.

Different KAS configurations are therefore appropriate for different levels of system autonomy. While 1-KAS offers simplicity and minimal annotation effort, it is more sensitive to variations in tree morphology and environmental conditions, making it better suited for semi-automatic or human assisted harvesting scenarios, which is consistent with previous observations that reduced structural constraints increase sensitivity to visual and morphological variability [29]. Overall, the field validation confirms that integrating pose-based keypoint detection with structural bounding box information effectively translates visual perception improvements into tangible gains in harvesting performance, providing a clear

pathway toward reliable intelligent vibration harvesting systems, in agreement with the strong influence of excitation and structural response on harvesting efficiency reported in walnut dynamic studies [30].

4. Conclusions

This study proposes a monocular vision-based method for clamping-point detection in walnut vibration harvesting by integrating pose estimation with structural visual information. The method uses operator guided annotations to identify the clamping point on the trunk and introduces bounding box annotations for auxiliary structural regions to provide spatial constraints. Results show that structural context improves detection accuracy and stability, making the method more robust to variations in tree morphology and environmental conditions. The comparative analysis of model architectures reveals that advanced pose estimation models, combined with structure assisted annotations, enhance detection precision. Field validation demonstrates that structure assisted strategies lead to higher and more stable net harvesting rates, confirming the practical applicability of the method for vibration harvesting operations. This study provides a reliable and deployable solution for clamping-point determination using monocular RGB images, suitable for low cost, embedded harvesting platforms. Future work will focus on improving robustness to challenging tree morphologies and adverse visual conditions, such as severe occlusion and illumination variation, which were identified as limiting factors in field experiments. In addition, extending the proposed framework toward multi view perception and integrating closed loop visual feedback with harvesting control will be explored to further enhance adaptability and enable fully autonomous vibration harvesting.

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Abbreviations

The following abbreviations are used in this manuscript:

- 5-KAS Five-key-part annotation strategy
- 2-KAS Two-key-part annotation strategy
- 1-KAS Single-key-part annotation strategy

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